CS257 Linear and Convex Optimization
Homework 13

Due: December 23, 2020

December 14, 2020

1. Consider the following problem

$$\begin{aligned}
\min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f(\boldsymbol{x}) = x_1^2 + x_2^2 \\
\text{s.t.} \quad & (x_1-1)^2 + (x_2-1)^2 \le 1 \\
& (x_1-1)^2 + (x_2+1)^2 \le 1
\end{aligned}$$

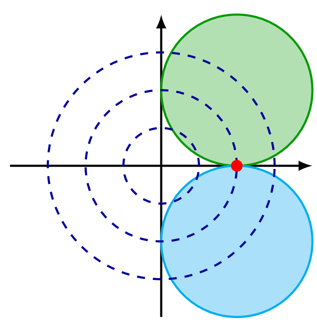
(a). Sketch the feasible set and the level set of the objective function. Find the optimal point $\boldsymbol{x}^*$ and the optimal value f^*.

(b). Find the Lagrange dual function and the dual problem. Find the dual optimal value ϕ^*. Does strong duality hold?

(c). Does Slater's condition hold? What can you conclude about the necessity of Slater's condition for strong duality?

(d). Is the dual optimal value ϕ^* attained by any dual feasible point? Do there exist any Lagrange multipliers satisfying the KKT conditions at $\boldsymbol{x}^*$?

Solution. (a). The only feasible point is $\boldsymbol{x}^* = (1,0)^T$, shown by the red dot. The dashed circles are the level sets of the objective function. The optimal value is $f^* = f(\boldsymbol{x}^*) = 1$.



(b). The Lagrangian is

$$\begin{aligned}
\mathcal{L}(\boldsymbol{x},\boldsymbol{\mu}) &= x_1^2 + x_2^2 + \mu_1[(x_1-1)^2 + (x_2-1)^2 - 1] + \mu_2[(x_1-1)^2 + (x_2+1)^2 - 1] \\
&= (1+\mu_1+\mu_2)x_1^2 - 2(\mu_1+\mu_2)x_1 + (1+\mu_1+\mu_2)x_2^2 - 2(\mu_1-\mu_2)x_2 + \mu_1 + \mu_2
\end{aligned}$$

If $\mu_1 + \mu_2 \le -1$, the paraboloid opens down, so

$$\phi(\boldsymbol{\mu}) = \inf_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}) = -\infty$$

If $\mu_1 + \mu_2 > -1$, the paraboloid opens up, and $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu})$ is minimized by

$$\begin{cases} x_1 = \frac{\mu_1+\mu_2}{1+\mu_1+\mu_2} \\ x_2 = \frac{\mu_1-\mu_2}{1+\mu_1+\mu_2} \end{cases}$$

so

$$\phi(\boldsymbol{\mu}) = \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2}$$

Thus the dual function is

$$\phi(\boldsymbol{\mu}) = \begin{cases} \frac{\mu_1+\mu_2-(\mu_1-\mu_2)^2}{1+\mu_1+\mu_2}, & \text{if } \mu_1 + \mu_2 > -1 \\ -\infty, & \text{if } \mu_1 + \mu_2 \le -1 \end{cases}$$

The dual problem is

$$\begin{aligned} \max_{\boldsymbol{\mu} \in \mathbb{R}^2} \quad & \phi(\boldsymbol{\mu}) = \begin{cases} \frac{\mu_1+\mu_2-(\mu_1-\mu_2)^2}{1+\mu_1+\mu_2}, & \text{if } \mu_1 + \mu_2 > -1 \\ -\infty, & \text{if } \mu_1 + \mu_2 \le -1 \end{cases} \\ \text{s.t.} \quad & \boldsymbol{\mu} \ge \mathbf{0} \end{aligned}$$

which is equivalent to

$$\begin{aligned} \max_{\boldsymbol{\mu} \in \mathbb{R}^2} \quad & \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2} \\ \text{s.t.} \quad & \boldsymbol{\mu} \ge \mathbf{0} \end{aligned}$$

By weak duality,

$$\phi^* \le f^* = 1.$$

When $\mu_1 = \mu_2 = \mu \to \infty$,

$$\phi(\mu, \mu) = \frac{2\mu}{1 + 2\mu} \to 1,$$

so

$$\phi^* = \sup_{\boldsymbol{\mu} \ge \mathbf{0}} \phi(\boldsymbol{\mu}) = 1 = f^*$$

Strong duality holds.

(c). Slater's condition does not hold, as both constraints hold with equality at the unique feasible point $\boldsymbol{x}^*$. This shows that Slater's condition is not necessary for strong duality to hold.

(d). Note

$$\phi(\boldsymbol{\mu}) = \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2} = 1 \implies (\mu_1 - \mu_2)^2 = -1$$

which is impossible, so the dual optimal value ϕ^* is not attained by any dual feasible point.

There are no Lagrange multipliers satisfying the KKT conditions at $\boldsymbol{x}^*$. Since the primal problem is convex, the satisfiability of KKT conditions would require the dual optimal value to be attainable.

$\square$

2. Consider the following minimization problem

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f(\boldsymbol{x}) = \begin{cases} x_1^3 + x_2^3, & \text{if } \boldsymbol{x} \geq \boldsymbol{0} \\ +\infty, & \text{otherwise} \end{cases} \\ \text{s.t.} \quad & x_1 + x_2 \geq 1 \end{aligned} \tag{P1}$$

Note the domain of f is $\operatorname{dom} f = \{\boldsymbol{x} \in \mathbb{R}^2 : \boldsymbol{x} \geq \boldsymbol{0}\}$ and the domain of the problem is $D = \operatorname{dom} f$.

(a). Since D is not the entire space, the dual function of this problem is defined by

$$\phi(\mu) = \inf_{\boldsymbol{x}\in D} \{f(\boldsymbol{x}) + \mu(1 - x_1 - x_2)\}$$

Find the explicit expression of $\phi(\mu)$.

(b). Find the dual optimal solution.

(c). What is the primal optimal value? Hint: Note f is convex on its domain.

(d). Note the primal problem (P1) is equivalent to

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f_1(\boldsymbol{x}) = x_1^3 + x_2^3 \\ \text{s.t.} \quad & x_1 + x_2 \geq 1 \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned} \tag{P2}$$

What's the dual function of this equivalent problem (P2)? Does strong duality hold for (P2)?

Remark. Equivalent primal problems can have very different dual problems. Not all dual problems are equally useful.

Solution. (a). Since D is not the entire space, the dual function of this problem is defined by

$$\begin{aligned} \phi(\mu) &= \inf_{x_1\geq 0, x_2 \geq 0} \{x_1^3 + x_2^3 + \mu(1 - x_1 - x_2)\} \\ &= \inf_{x_1 \geq 0}\{x_1^3 - \mu x_1\} + \inf_{x_2\geq 0}\{x_2^3 - \mu x_2\} + \mu \\ &= \begin{cases} -4(\frac{\mu}{3})^{3/2} + \mu, & \text{if } \mu \geq 0 \\ \mu, & \text{if } \mu \leq 0 \end{cases} \end{aligned}$$

(b). For $\mu > 0$,

$$\phi''(\mu) = -\frac{1}{3}\sqrt{\frac{3}{\mu}} < 0$$

so ϕ is concave on $(0, +\infty)$.

$$\phi'(\mu) = 1 - 2\sqrt{\frac{\mu}{3}} = 0 \implies \mu = \frac{3}{4}$$

So

$$\max_{\mu>0} \phi(\mu) = \phi(\frac{3}{4}) = \frac{1}{4}$$

and

$$\phi^* = \max_{\mu\geq 0} \phi(\mu) = \max\left\{\phi(0), \phi(\frac{3}{4})\right\} = \frac{1}{4}$$

(c). Slater's condition holds. For example, $\boldsymbol{x} = (1,1) \in \text{int}D$ and $1 - x_1 - x_2 < 0$. So $f^* = \phi^* = \frac{1}{4}$.

(d). The dual function is

$$\phi_1(\boldsymbol{\mu}) = \inf_{\boldsymbol{x}} \{x_1^3 + x_2^3 - \mu_1(x_1 + x_2 - 1) - \mu_2 x_1 - \mu_3 x_3\} = -\infty$$

So the dual optimal value is $\phi_1^* = -\infty \neq f_1^* = f^* = \frac{1}{4}$. Strong duality does not hold for (P2).

$\square$